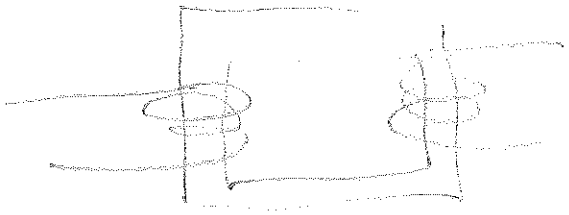
Practice Problems
Calculus Physics II

Name: Key **Date:** _____

Chapter 49 – Induction in Circuits

1) Which principle does best explain how a transformer works?

A) Self inductance
B) Motional emf
C) Mutual inductance
D) Magnetic flux
E) Eddy currents



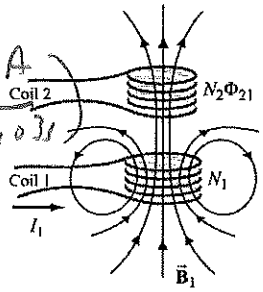
2 coils of wire

2) Two coils of wire are next to each other as shown, coil 1 has 10 loops and coil 2 has 20 loops. A current in coil 1 starts from zero and reaches a maximum value of 8 A in 0.03 s, if the mutual inductance of the two coils is 12 mH, what is the induced potential in the second coil?

$M = 12 \times 10^{-3}\ H$

$V_{ind} = -M\left(\frac{\Delta I_p}{\Delta t}\right) = -(12 \times 10^{-3}\ H)\left(\frac{8\ A}{0.03\ s}\right)$

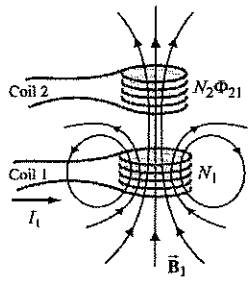
$V_{ind} = -3.2\ V$



3) Two coils of wire are next to each other as shown, coil 1 has 100 loops and coil 2 has 400 loops. A current in coil 1 starts from zero and reaches a maximum value of 2 A in 4 x 10^{-5} s, what is the mutual inductance of the system of coils if the induced voltage in the second coil is 6 V?

$V_{ind} = -M\left(\frac{\Delta I_p}{\Delta t}\right)$ Solve for M

$M = \frac{\Delta t\, V_{ind}}{\Delta I_p} = 1.2 \times 10^{-4}\ H$



4) A transformer actually transforms

A) non-safe forms of energy to safe forms of energy.
B) voltage.
C) magnetic field lines.
D) generators into motors.
E) All of the above choices are correct.

5) The output power of an ideal transformer is

A) greater than the input power.
B) equal to the input power.
C) smaller than the input power.
D) may be any of the above

6) The primary coil of an ideal transformer has 100 turns and its secondary coil has 400 turns. If the ac voltage applied to the primary coil is 120 V, what voltage is present in its secondary coil?

$\frac{V_S}{V_P} = \frac{N_S}{N_P}$ | $V_S = \left(\frac{N_S}{N_P}\right) V_P = 480\,V$

7) In an ideal transformer, how many turns are necessary in a 120 V primary if the 20 V secondary has 100 turns?

$\frac{V_S}{V_P} = \frac{N_S}{N_P}$ | $N_P = \left(\frac{V_P}{V_S}\right) N_S = 600$ turns

8) The primary coil of a transformer has 1000 windings of wire, and the secondary has 150 windings of wire, if the secondary uses 200 W of power, what is the power required by the primary.

$P_P = P_S = 200\,W$

9) A transformer has $N_P = 100$ loops on the primary and $N_S = 20$ loops on the secondary, if the secondary's voltage is $V_S = 20$ V and there is a current of $I_S = 4$ A through the secondary, what is the <u>current</u> through the primary coils?

$\frac{N_S}{N_P} = \frac{I_P}{I_S}$ | $I_P = I_S\left(\frac{N_S}{N_P}\right) = 0.8A$

10) The power used by the primary side of a transformer is 500 W and has a current of 2 A flowing with $N_P = 100$ loops of wire, if the secondary has $N_S = 50$ loops of wire, what is the volage across the secondary?

$P_P = I_P V_P$

$V_P = \frac{P_P}{I_P} = 250V$ | $\frac{N_S}{N_P} = \frac{V_S}{V_P}$ | $V_S = \left(\frac{N_S}{N_P}\right) V_P = 125V$

11) The power used by the primary side of a transformer that is used to charge a calculator is 12 W and has a current of 0.1A flowing with $N_P = 600$ loops of wire, if the secondary has 2 A of current flowing through it,

A) What is the volage across the secondary?

$P_S = 12W = I_S V_S$ | $V_S = \frac{P_S}{I_S} = \frac{12W}{2A} = 6V$

B) How many coils of wire are on the secondary?

$P_P \frac{V_S}{V_P} = \frac{I_P}{I_S} = \frac{N_S}{N_P}$ | $N_S = N_P\left(\frac{I_P}{I_S}\right) = 30$ coils

12) Which of the following statements about inductors are correct?

A) When an inductor and a resistor are connected in series with a DC battery, the current in the circuit is constant over time.
B) When an inductor and a resistor are connected in series with a DC battery, the current in the circuit is zero after a very long time.
C) Inductors store energy by building up charge.
D) When it is connected in a circuit, an inductor always resists having current flow through it.
(E) An inductor always resists any change in the current through it.

13) Disconnect a small-voltage battery from a coil of many loops of wire and a large voltage may be produced by

A) the sudden collapse in the magnetic field.
B) latent energy in the battery.
C) the resistance of the battery to a change in polarity.
D) electrons already in the wire.
E) the electric field between the battery terminals.

14) The voltage induced in self-inductions is

A) always in a direction opposing the changing voltage that produced it.
B) likely higher than the changing voltage that causes it.
C) both of these
D) neither of these

15) What is the inductance of an iron filled solenoid (μ_{iron} = 0.25 Tm/A) 30 cm long, with 100 turns of wire and a cross-sectional area of 1 x 10^{-4} m^2?

$L = \frac{\mu_I N^2 A}{\ell} = 0.83\ H$

16) A resistor and an inductor are connected in series to an ideal battery of constant terminal voltage. At the moment contact is made with the battery, the voltage across the inductor is

A) greater than the battery's terminal voltage.
B) equal to the battery's terminal voltage.
C) less than the battery's terminal voltage, but not zero.
D) zero.
E) equal to the voltage across the resistor.

Larger $\Delta\Phi$ → Larger V_{emf}

17) A coil with a self-inductance of 6.0 H is connected to a dc source through a switch. As soon as the switch is closed at t = 0 s, the rate of change of current is 2.0 A/s. What is the emf induced in this coil at t = 0 s?

$V_{em} = -L\left(\frac{\Delta I}{\Delta t}\right) = -(6\ H)(2\ A/s) = -12\ V$

18) The current flowing through a circuit is changing at a rate of 6.0 A/s. If the circuit contains an inductor and the voltage across the inductor is 24 V, what is the inductance of the inductor?

$V_{em} = -L\left(\frac{\Delta I}{\Delta t}\right)$ | $L = \frac{V_{em}}{\left(\frac{\Delta I}{\Delta t}\right)} = \frac{24\ V}{6\ A/s}$

$L = 4\ H$

19) In the figure, the current in a solenoid having no appreciable resistance is flowing from *b* to *a* and is decreasing at a rate of 9.6 A/s. The self-induced V_{emf} in the solenoid is found to be 8.4 V.



A) Determine the self-inductance (L) of the solenoid?

$V_{ens} = L\left(\frac{\Delta I}{\Delta t}\right)$ | $L = \frac{V_{ens}\,\Delta t}{\Delta I}$ = 0.875 H

B) Which point, *a* or *b* is at higher potential?

Vens adds to curnt because current is decreasing | Vens + | − a

20) Using Figure A, an inductor and a fully charged capacitor are wired in series with a switch. The "a" side of the capacitor has an initial positive charge. Where is the electrical energy stored at t = 0 s?

A) Magnetic Field.
B) Electric Field
C) Current in circuit.
D) Voltage in circuit.
E) The Capacitance of the capacitor.
F) The inductance of the Inductor.

Figure A



21) Using Figure A, after the switch is closed and the charge on the capacitor is zero, where is the electrical energy stored?

A) Magnetic Field.
B) Electric Field
C) Current in circuit.
D) Voltage in circuit.
E) The Capacitance of the capacitor.
F) The inductance of the Inductor.

22) Using Figure A, after the inductor discharges for the first time and the capacitor is fully re-charged, what type of charge is on side "a' and side "b" of the capacitor?

A) "a" has a positive charge.
B) "a" has a negative charge.
C) "a" is neutral, there is no charge on "a".
D) The type of charge cannot be determined.

23) Using Figure A, how does the total electrical energy of the circuit change as the capacitor is charged and discharged?

A) The total energy remains in the electric field.
B) The total energy remains in the magnetic field.
C) The total energy remains zero.
D) The total energy does not change with time.

KEY

1) C
2) -3.2 V
3) 1.2×10^{-4} H
4) B
5) B
6) 480 V
7) 600 loops
8) 200 W
9) 0.8 A
10) 125 V
11) A) 6 V B) 30 coils
12) E
13) A
14) A
15) 0.83 H
16) B
17) -12 V
18) 4 H
19) A) 0.875 H B) point "a"
20) B
21) A
22) B
23) D